

Application of Gauss's Law

Symmetrical
Charge
Distributions

To determine $\mathbf{D}_S$ for
known charge
distributions

Asymmetrical
Charge
Distribution

To determine function of
 $\mathbf{D}$ and eventually finding
Maxwell's first equation

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Application of Gauss's Law

Symmetrical Charge Distributions

For a known charge distribution, finding D_S using Gauss's Law.

$$\oint_S \mathbf{D}_S \cdot d\mathbf{S} = Q_{encl}$$

- This is an example of an integral equation in which the unknown quantity lies inside the integral.
- The solution is easy if we are able to choose a **Gaussian surface** which satisfies two conditions:
 - 1) $\mathbf{D}_S$ is everywhere either normal or tangential to the closed surface, so that $\mathbf{D}_S \cdot d\mathbf{S}$ becomes either $D_S dS$ or zero, respectively.
 - 2) On that portion of the closed surface for which $\mathbf{D}_S \cdot d\mathbf{S}$ is not zero, $D_S = \text{constant}$
- Then, D_S can be brought out of the integral and $\int_S dS$ can be solved.

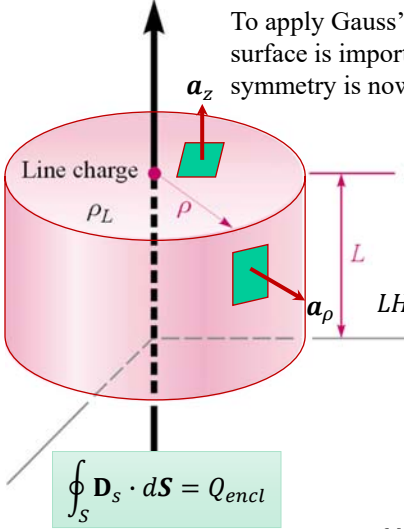
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Gauss's Law: Symmetrical Charge Distributions

Gauss's law simplifies the task of finding **D** near an infinite line charge.

To apply Gauss's law, finding the closed surface / *Gaussian* surface is important as discussed and for that applying symmetry is now a **must** condition.



1. With which coordinates does the field vary (or of what variables is *D* a function)?

2. Which components of **D** are present?

$\mathbf{D} = D_\rho \mathbf{a}_\rho$

$D_\rho = f(\rho)$

$$\begin{aligned} LHS &= \oint_{cylinder} \mathbf{D}_s \cdot d\mathbf{S} = \int_{sides} + \int_{top} + \int_{bottom} \\ &= \int_{sides} D_\rho \mathbf{a}_\rho \cdot dS_\rho \mathbf{a}_\rho + \int_{top} D_\rho \mathbf{a}_\rho \cdot dS_z \mathbf{a}_z \\ &\quad + \int_{bottom} D_\rho \mathbf{a}_\rho \cdot dS_{-z} (-\mathbf{a}_z) \end{aligned}$$

$$\oint_S \mathbf{D}_s \cdot d\mathbf{S} = Q_{encl}$$

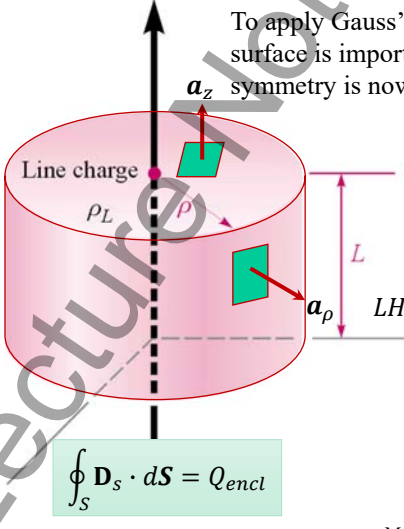
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1. With which coordinates does the field vary (or of what variables is *D* a function)?

2. Which components of **D** are present?

$\mathbf{D} = D_\rho \mathbf{a}_\rho$

$D_\rho = f(\rho)$

$$\begin{aligned} LHS &= D_\rho \int_{sides} dS_\rho + 0 \int_{top} dS_z + 0 \int_{bottom} dS_{-z} \\ &= D_\rho \int_{z=0}^L \int_{\phi=0}^{2\pi} \rho d\phi dz \end{aligned}$$

$$\oint_S \mathbf{D}_s \cdot d\mathbf{S} = Q_{encl}$$

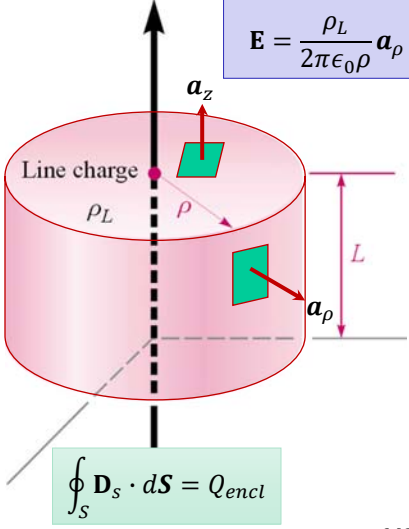
$$LHS = D_\rho 2\pi \rho L = Q_{encl} \text{ (RHS)}$$

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Gauss's Law: Symmetrical Charge Distributions

Gauss's law simplifies the task of finding **D** near an infinite line charge.



$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho$$
$$\mathbf{D} = D_\rho \mathbf{a}_\rho$$
$$D_\rho = f(\rho)$$
$$LHS = D_\rho 2\pi\rho L = Q_{encl}$$
$$RHS = Q_{encl} = \int_L \rho_L dL$$
$$Q_{encl} = \int_{z=0}^L \rho_L dz = \rho_L L$$
$$D_\rho 2\pi\rho L = \rho_L L$$
$$D_\rho = \frac{\rho_L}{2\pi\rho}$$
$$\mathbf{D} = D_\rho \mathbf{a}_\rho$$
$$\mathbf{D} = \frac{\rho_L}{2\pi\rho} \mathbf{a}_\rho = \epsilon_0 \mathbf{E}$$

$$\oint_S \mathbf{D}_s \cdot d\mathbf{S} = Q_{encl}$$

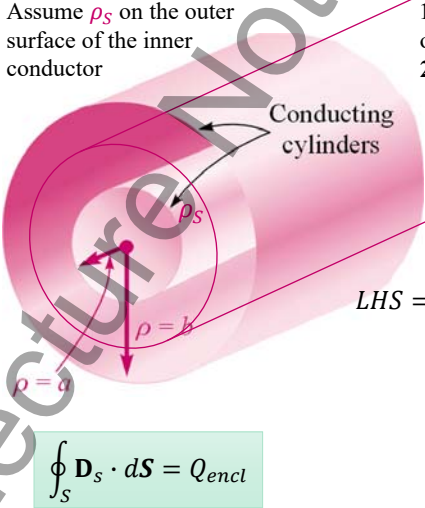
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Gauss's Law: Infinite Coaxial Cable

Infinite coaxial cable: A problem extremely difficult to solve using Coulomb's Law

Assume ρ_s on the outer surface of the inner conductor



1. With which coordinates does the field vary (or of what variables is **D** a function)?

2. Which components of **D** are present?

$$\mathbf{D} = D_\rho \mathbf{a}_\rho$$
$$D_\rho = f(\rho)$$

A right circular cylinder of length L and radius ρ , where $a < \rho < b$, is necessarily chosen as the gaussian surface

$$LHS = D_\rho \int_{sides} dS_\rho + 0 \int_{top} dS_z + 0 \int_{bottom} dS_{-z}$$
$$= D_\rho \int_{z=0}^L \int_{\phi=0}^{2\pi} \rho d\phi dz$$
$$LHS = D_\rho 2\pi\rho L = Q_{encl} (RHS)$$

$$\oint_S \mathbf{D}_s \cdot d\mathbf{S} = Q_{encl}$$

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EXAMPLE 3.2

Let us select a 50-cm length of coaxial cable having an inner radius of 1 mm and an outer radius of 4 mm. The space between conductors is assumed to be filled with air. The total charge on the inner conductor is 30 nC. We wish to know the charge density on each conductor, and the **E** and **D** fields.

Solution. We begin by finding the surface charge density on the inner cylinder,.

$$\rho_{S, \text{inner cyl}} = \frac{Q_{\text{inner cyl}}}{2\pi aL} = \frac{30 \times 10^{-9}}{2\pi (10^{-3})(0.5)} = 9.55 \mu\text{C/m}^2$$

The negative charge density on the inner surface of the outer cylinder is

$$\rho_{S, \text{outer cyl}} = \frac{Q_{\text{outer cyl}}}{2\pi bL} = \frac{-30 \times 10^{-9}}{2\pi (4 \times 10^{-3})(0.5)} = -2.39 \mu\text{C/m}^2$$

The internal fields may therefore be calculated easily:

$$D_\rho = \frac{a\rho_S}{\rho} = \frac{10^{-3}(9.55 \times 10^{-6})}{\rho} = \frac{9.55}{\rho} \text{ nC/m}^2$$
$$E_\rho = \frac{D_\rho}{\epsilon_0} = \frac{9.55 \times 10^{-9}}{8.854 \times 10^{-12}\rho} = \frac{1079}{\rho} \text{ V/m}$$

Both of these expressions apply to the region where $1 < \rho < 4$ mm. For $\rho < 1$ mm or $\rho > 4$ mm, **E** and **D** are zero.

D3.5.

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